

BOM CHECK (automated - verify before acting)

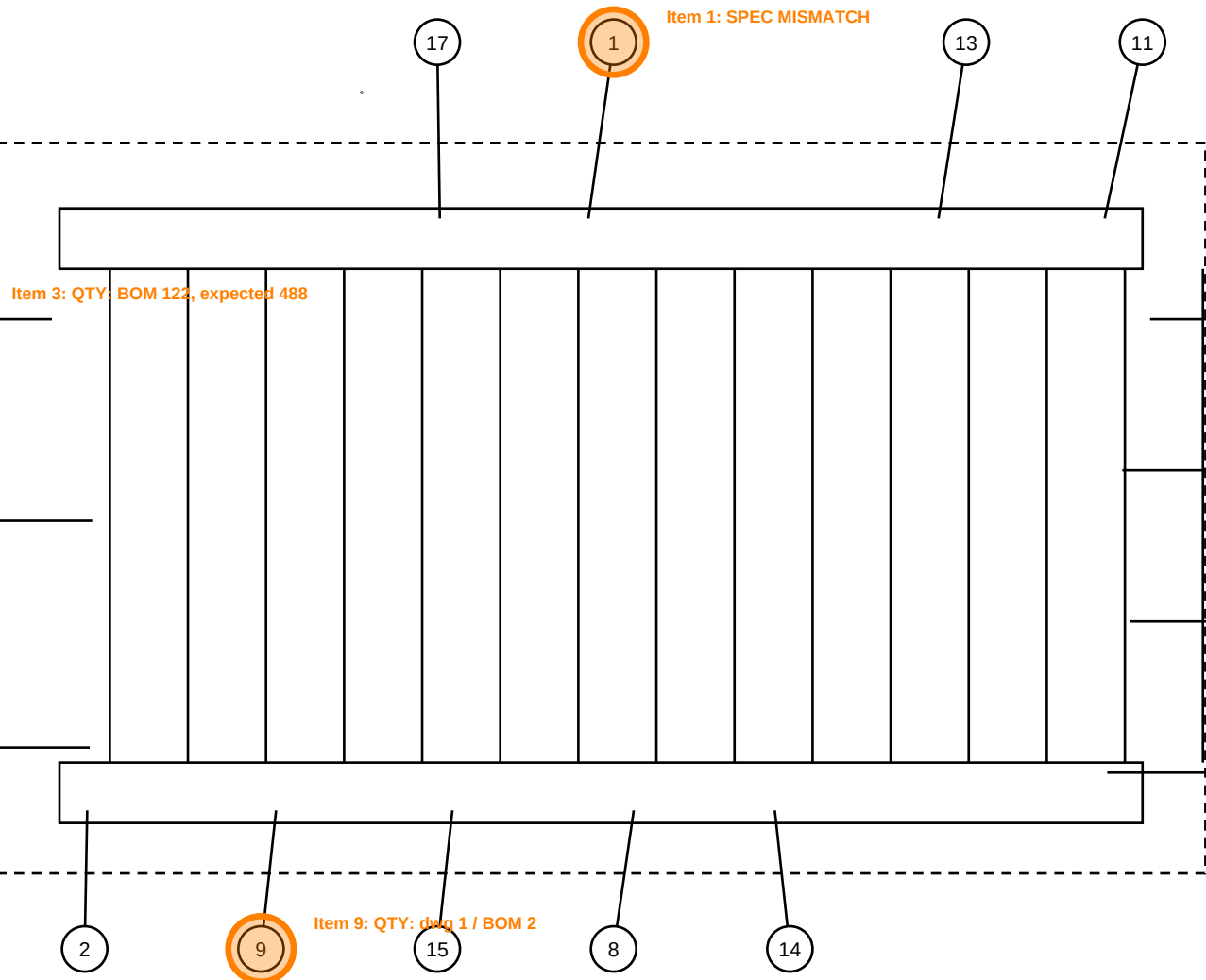
- red = on drawing, missing in BOM
- orange = qty / spec / revision mismatch
- yellow = confirm match / extra / duplicate
- blue = suggestion (implied item, rule-based)

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Item 1: SPEC MISMATCH

Item 3: QTY: BOM 122, expected 488

Item 9: QTY: dwg 1 / BOM 2

BOM LACKS: Cladding

Item 10: MISSING IN BOM

Pos	Description	Material Spec	Qty
1	INLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B	1
2	OUTLET HEADER 219.1 OD X 8.0 THK	SA 106 GR.B	1
3	ECONOMISER TUBE 38.1 OD X 3.6 THK	SA 210 GR.A1	122
4	RETURN BEND 38.1 OD	SA 210 GR.A1	48
5	TUBE SUPPORT PLATE 12 THK	IS 2062 GR.B	13
6	INLET FLANGE 200 NB CL.150	SA 105	1
7	OUTLET FLANGE 200 NB CL.150	SA 105	1
8	DRAIN VALVE 25 NB	SA 105	2
9	VENT VALVE 15 NB	SA 105	1
10	THERMOWELL 1/2 NPT	SS 316	2
11	PRESSURE TAPPING 1/2 NB	SA 105	2
12	STUD BOLT M20 X 90 WITH 2 NUTS	SA 193 B7	16
13	LIFTING LUG 16 THK	IS 2062 GR.B	4
14	CASING PLATE 3 THK	IS 2062	17
15	EXPANSION JOINT 600 X 600	FABRIC	1
16	NAME PLATE	SS 304	1
17	INSPECTION DOOR 300 X 300	IS 2062	2

NOTES:

1. ALL DIMENSIONS ARE IN MM.
2. ALL WELDING AS PER IBR / ASME SEC I.
3. INSULATE WITH 100 THK MINERAL WOOL AND CLAD WITH 0.6 THK ALUMINIUM SHEET.
4. PAINT: ONE COAT RED OXIDE PRIMER.
5. TUBE QTY IS PER BANK. TYPICAL FOR 4 BANKS.

ECONOMISER ASSEMBLY

DRG NO : EC-02-GA-001

DRG REV : B

SCALE : NTS SIZE : A3

SAMPLE (SYNTHETIC) DRAWING

BOM CHECK SUMMARY

Drawing : sample_02_drawing.pdf (Rev B)
BOM : sample_02_bom.xlsx (Rev A)
Input : vector PDF
RUN RELIABILITY : HIGH

Drawing items checked : 17
BOM lines read : 18
BOM completeness : 88.2% (drawing items that have a BOM line)

Findings: implied 1, multiplier 1, omission 2, quantity 1, revision 1, spec 1

[RED] Item 10 'THERMOWELL 1/2 NPT' (qty 2) is on the drawing but has no BOM line
[RED] Item 12 'STUD BOLT M20 X 90 WITH 2 NUTS' (qty 16) is on the drawing but has no BOM line
[ORANGE] Item 1: specification mismatch - dimension: drawing ['8'] vs BOM ['10']
[ORANGE] Item 3: drawing note says qty is per bank, x4; BOM has 122 but total should be 488
[ORANGE] Item 9: quantity differs (drawing 1, BOM 2)
[ORANGE] Revision mismatch: drawing is Rev B, BOM is Rev A
[BLUE] Suggestion (Cladding): the drawing notes imply it, but no matching line was found in the BOM

Automated check - not a substitute for engineering review. Blue items are rule-based suggestions.